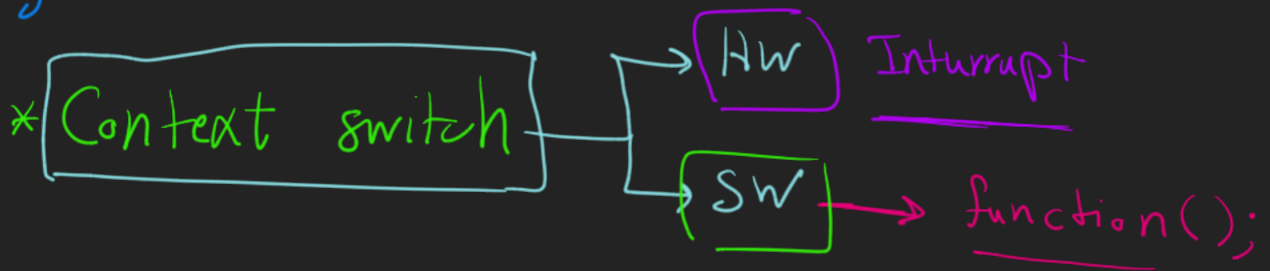
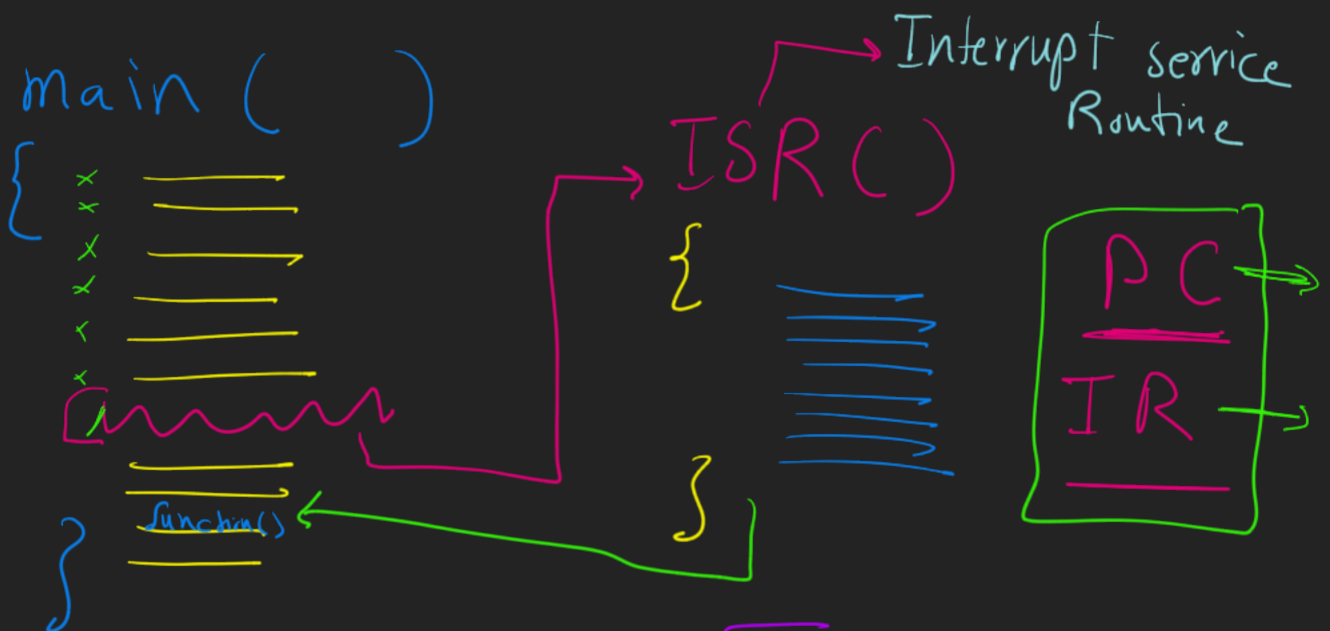


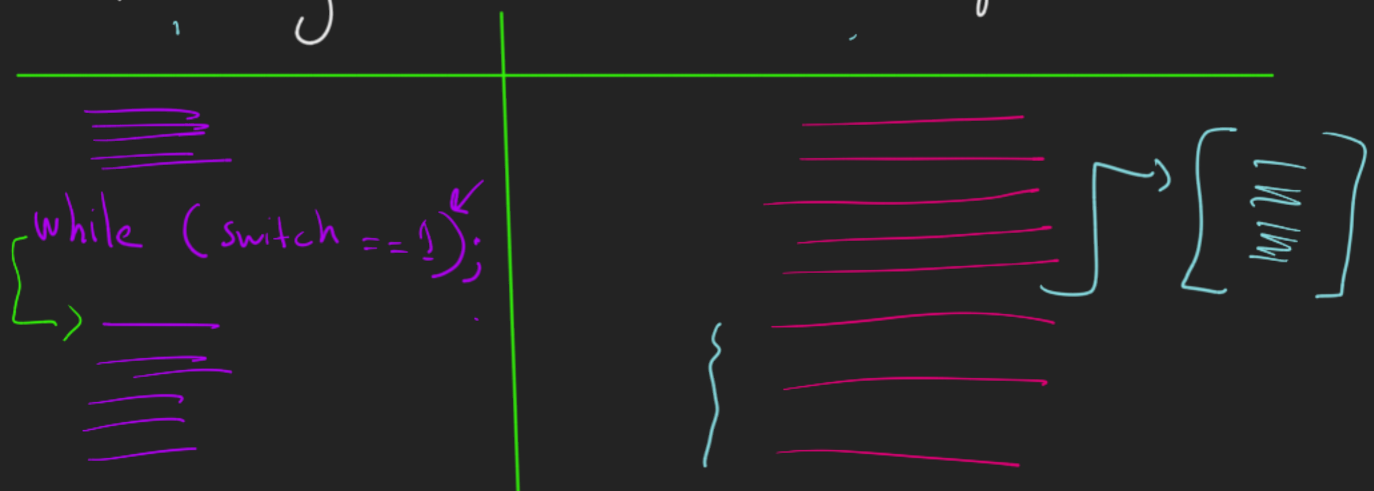
* Interrupt | Airbag | Fire system



Interrupt / ? → signal / Didn't response
 ↳ Response



* Polling Vs Interrupt



① PIF peripheral Interrupt Flag

② PIE

③ GIE Global

Enable 1
Disable 0

Enable 1
Disable 0

PIF	PIE	GIE	INT
0	0	0	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Interrupt

→ Maskable

I can escape from it.

ADC, Timer, SPI
PIE

→ Non Maskable → Reset

Action I can't Ignore.

HW Circuit

GIE → 1
→ 0

Static Flash

Dynamic RAM

Vector table

Vector 1	Address
Vector 2	Address
Vector 3	Address

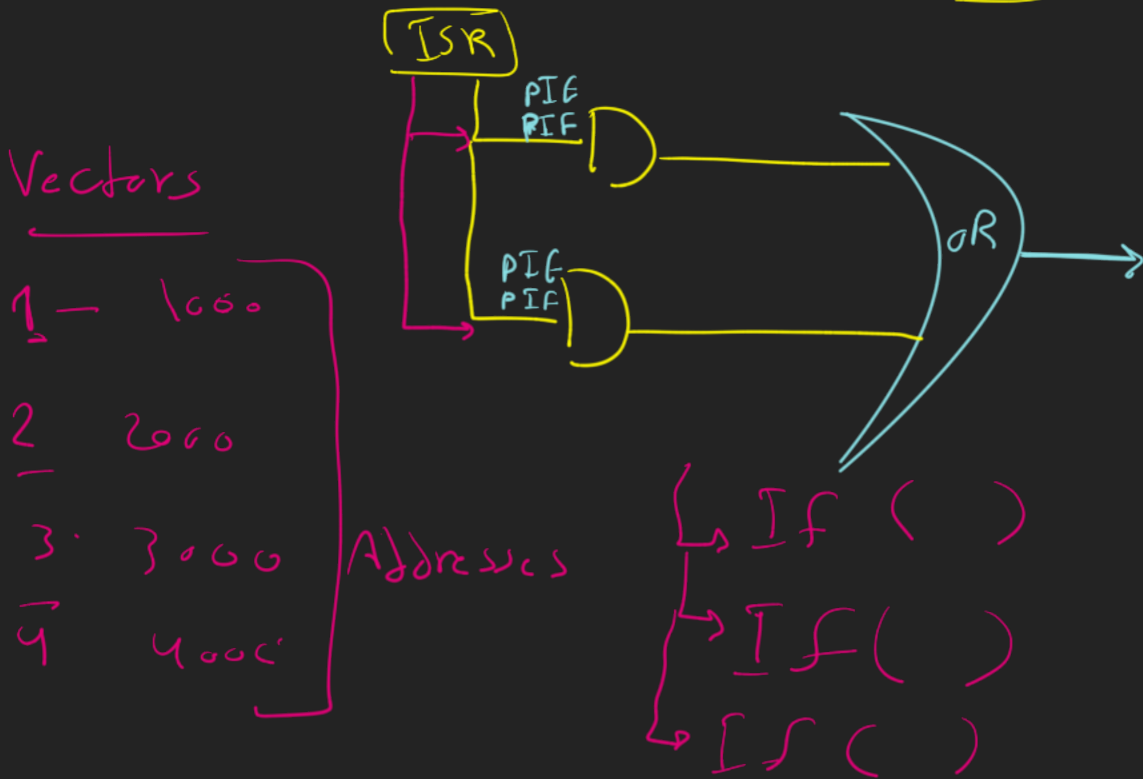
PIC → Programmable Int Counter

Interrupt	ID
ADC	1
Timer	2
UART	3
SPI	4
Memory Watch	

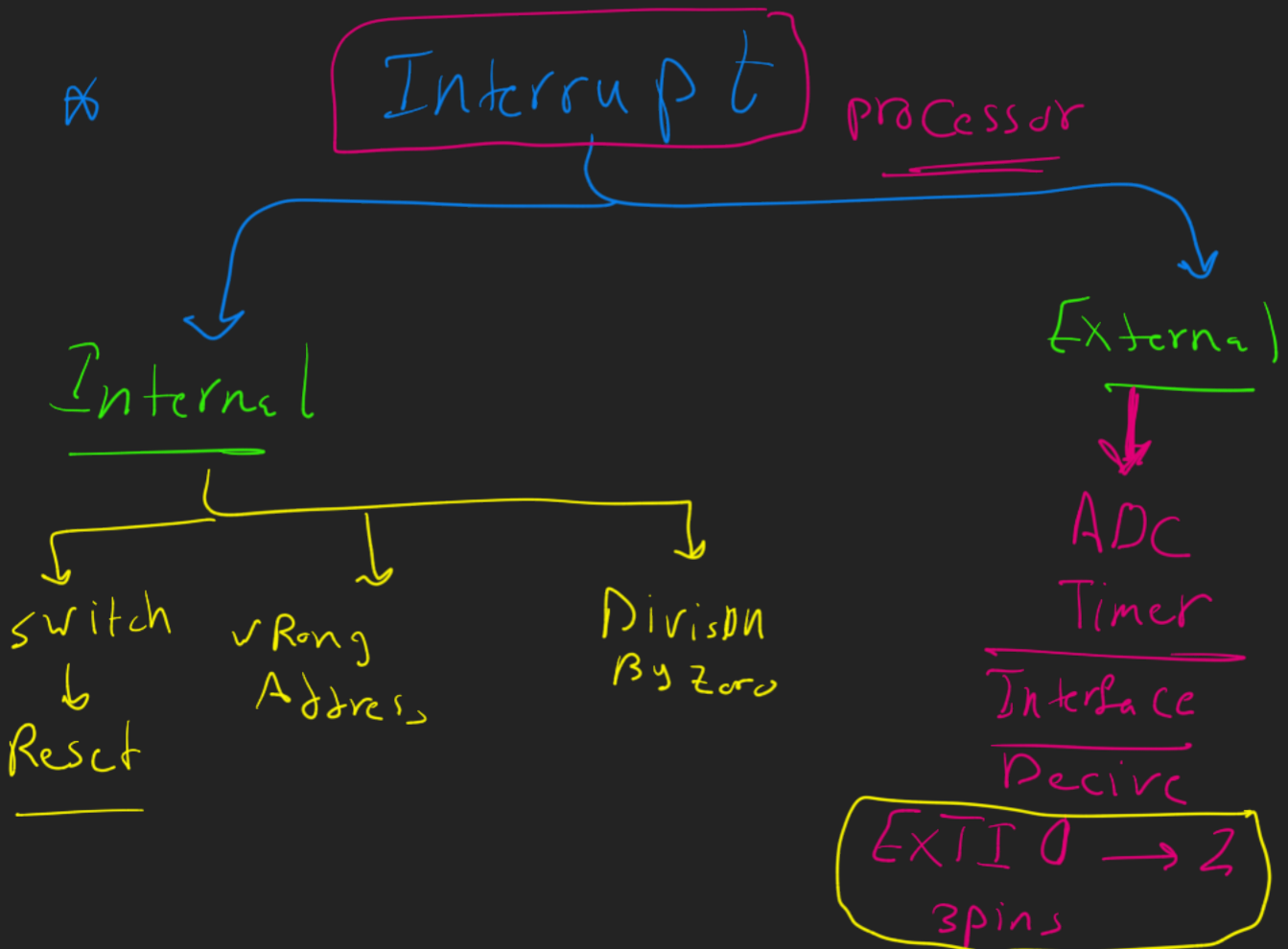
→ processor

* I can't change

Dynamic Vector table SW



time of Execution is late HW



* Interrupt time Diagram

Air bag → Turn Airbag in time ?

$(\underline{20} - \underline{50}) \text{ ms}$

Interrupt time $> (20 - 50)$

Compressor

→ Airbag

time

Normal Code

* Interrupt happens Flag

1 MSec

20ms

Context switch

Interrupt Code (execution time)

Context switch

* Normal Code

* External Interrupt

EXTI0, EXTI1, EXTI2

* Trigger

1 → falling edge

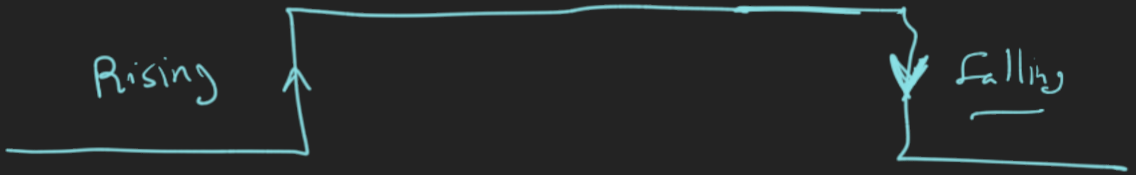
2 → low level

3 → Rising edge

4 → Any logical change (1,3)



20 cycle



* clear flag

① HW → Int

② SW → write 0 to it

③ SW → // 1 on it

④ SW → Read flag

* SW Interrupt

Function (); with time to execute

